

The Pressure Wall

Concept Document

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Pitch Line

"The Pressure Wall" — a high-stakes 2D platformer where you run, jump, and parkour your way to the exit gate before a relentless wall crushes you.

Introduction

The Pressure Wall is a 2D single-player platformer that blends fast-paced action with intense parkour challenges in a world where danger is ever-present. Players control a character who must navigate a series of obstacles to reach a gate marking the final goal of the level. The game's defining feature is a relentless, ever-advancing wall that pursues the player — if it catches up, it's game over, creating a constant sense of urgency and pressure. Moreover, the wall accelerates whenever the player looks left or remains still, encouraging continuous movement and split-second decisions. To further challenge progression, players must also avoid deadly spikes, demanding precision and careful timing in every parkour maneuver.

Demographic Breakdown

Despite its simple controls, the game requires strategy, quick reflexes, and precision. For this reason, it is recommended for ages 8 and up and is only available on PC.

Feature List

- 2D single-player platformer;
- Fast-paced action with high-precision parkour;
- Core mechanic: a constantly advancing pressure wall that forces forward movement;
- Instant game over upon touching the wall or spikes;
- The wall accelerates if the player remains idle or looks backward;

- Gameplay designed to encourage and reward continuous, agile movement;
- Constant tension and urgency-driven atmosphere;
- In-game pause menu available.

Feature List Breakdown

- **2D Single-Player Platformer:** A 2D platformer game focused on a single-player experience, with challenges designed for precise execution and quick, repetitive attempts.
- **Fast-Paced, Precision-Based Gameplay:** High-speed action that demands quick reflexes, precise control, and mastery of parkour movements to survive and progress.
- **Constantly Advancing Pressure Wall:** A wall of energy advances relentlessly through the level, serving as the core game mechanic and forcing the player to maintain constant forward momentum.
- **Instant Death on Contact:** Any contact with the pressure wall or spikes results in immediate game over, emphasizing the importance of precision and timing.
- **Wall Acceleration:** The wall accelerates if the player stands still or looks back, penalizing hesitation and heightening the urgency of continuous advancement.
- **Movement-Driven Level Design:** Levels are built to encourage flow, agility, and movement combos, rewarding players who maintain consistent speed and control.
- **Tension-Focused Atmosphere:** The combination of a fast pace, punishing mechanics, and forced progression creates a constant atmosphere of tension and psychological pressure on the player.
- **Accessible Pause Menu:** The player can pause at any time to access basic options, ensuring accessibility without disrupting the overall pace of the experience.

Formal Description

Game State:

The game state encapsulates all relevant data at any point in time, including:

- `character_position`: The 2D coordinates of the player character (`CharacterBody2D`).

- `character_velocity`: The player's current velocity (Vector2).
- `wall_position`: The horizontal position of the moving wall (float).
- `wall_visible`: A boolean flag for the wall's visibility.
- `wall_opacity`: The wall's alpha value (float, between 0.0 and 1.0).
- `is_paused`: A boolean flag indicating if the game is paused.
- `game_result`: The final outcome ("win" / "lose").
- `death_reason`: Cause of death ("spikes", "wall", or empty if not applicable).

$S = \{\text{character_position}, \text{character_velocity}, \text{wall_position}, \text{wall_visible}, \text{wall_opacity}, \text{is_paused}, \text{game_result}, \text{death_reason}\}$

Game View:

The player has complete visibility of all relevant elements, with no hidden information. On screen, they can observe:

- Their character's current position within the level;
- Static obstacles (spikes);
- A moving wall that advances continuously;
- The final exit or gate;
- The pause menu, when activated, which includes:
 - The status indicator ("Game Paused");
 - Options to Resume, Restart, and Return to Main Menu.

Game Setup:

At the start of the level:

- `wall_visible` set to false
- `wall_opacity` set to 0
- `wall_position` starts off-screen
- `is_paused` set to false
- `game_result` set to undefined
- `death_reason` set to empty
- The pause menu is hidden

- The game is running normally (`get_tree().paused = false`)

Player Actions:

The available in-game actions are:

- **Character movement:**
Indirectly controls the wall speed. If the player stops or moves backward, the wall accelerates.
- **Pause/Resume the game:**
Using the `ui_cancel` command: `toggle_pause()`
- **Interacting with the pause menu:**
 - Resume the game;
 - Restart the level;
 - Return to the main menu.

`A = {move_character(Vector2), toggle_pause(), resume_game(), restart_level(), return_to_menu()}`

Victory Conditions:

The player wins by guiding the character to the final gate.

Formal win condition:

If the character collides with the final gate, then `game_result = "win"`.

The player loses if the character collides with any of the following hazards:

- **Spikes:**
`death_reason = "spikes"`
- **Moving wall:**
`death_reason = "wall"`

In both fatal collision cases, the game result will be: `game_result = "lose"`.

Progression of Play:

The game progression works as follows:

Each frame (in the `_process(delta)` method):

- If the game is not paused, the wall advances continuously;
- The player tries to move faster than the wall;

- Collisions are constantly checked:
 - Contact with **spikes** → defeat;
 - Contact with the **wall** → defeat;
 - Contact with the **gate** → victory.

When the game ends:

- The pause state is disabled;
- The result is stored in the global variable Global;
- The scene transitions to the results screen.